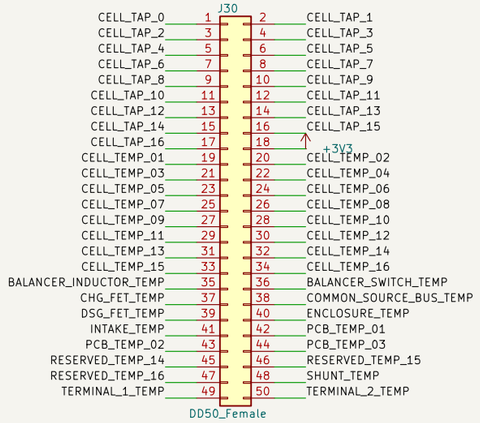
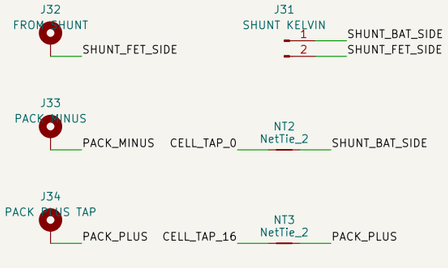
1. EXTERNAL SENSE & POWER INTERFACE

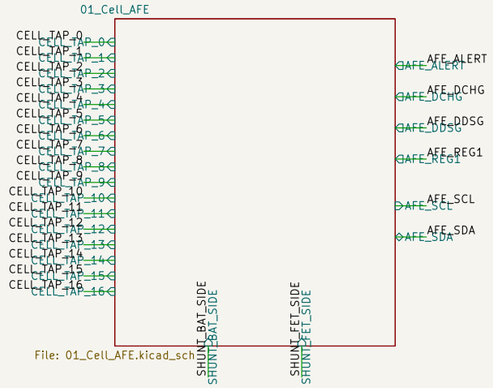
DD-50 Sense Harness, M8 Power Lugs & Kelvin Shunt Sense



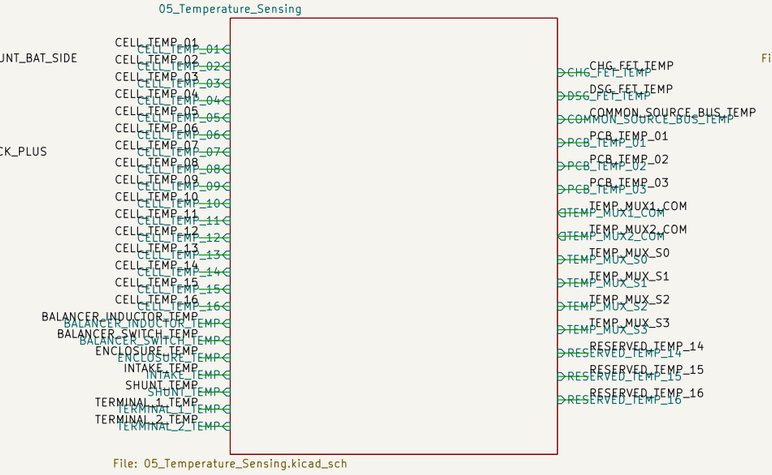


2. CELL MANAGEMENT & SENSING

BQ76952 AFE, Active Balancer Monitor & 32-Ch NTC Matrix

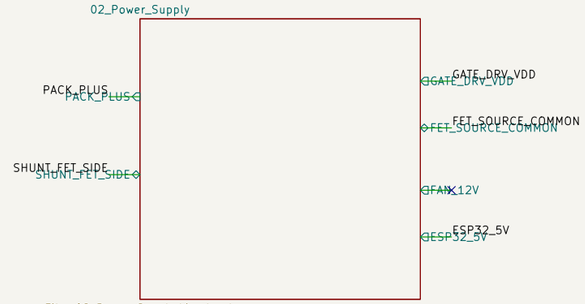


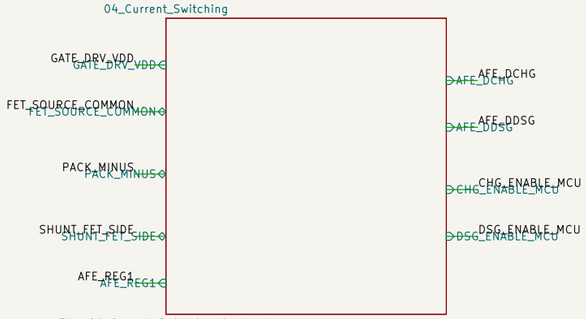




3. POWER CONVERSION & SWITCHING

Buck Regulators, Isolated Gate Drivers & 200A Cutoff Switches



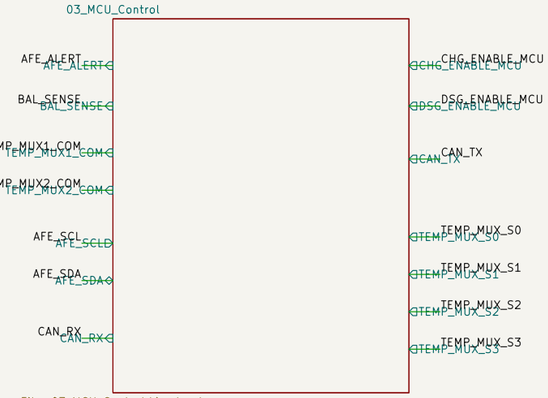


BMS ARCHITECTURE & GROUND RULES:

1. Low-Side Shunt: External 100A/75mV current shunt between battery negative and switches.
   Kelvin sense wires feed Sheet 01 (AFE) via J31.
2. Star Ground Tie: NetTie NT1 (Sheet 02) ties logic GND to SHUNT_FET_SIDE so BMS quiescent draw
   is measured by the shunt. NT2 and NT3 tie CELL_TAP_0 and CELL_TAP_16 at single copper points.
3. Galvanic Segregation: Logic GND, AFE VSS, and CAN ISO_GND remain strictly isolated.
   Never bridge GND, VSS, or CELL_TAP_0 at root level.

4. DIGITAL PROCESSING & ISOLATED TELEMETRY

CH32V203 RISC-V Controller, I2C2 FRAM & Isolated CAN Pack Link





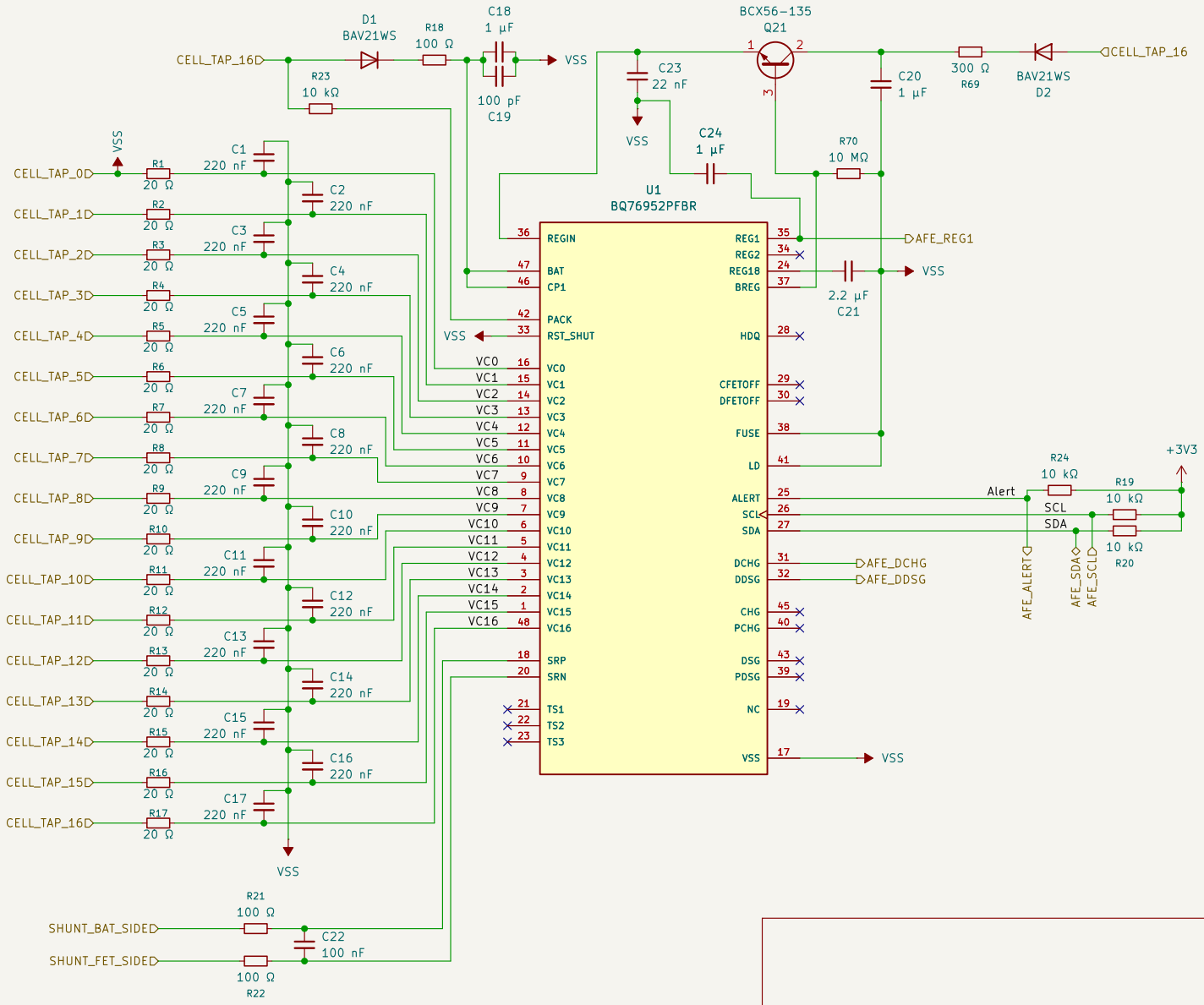
Designed for 16S 48V/51.2V 200A Continuous ESS / Electric Mobility Battery Systems
Galvanic Isolation: Logic GND, Power Shunt-Referenced GND, and Isolated CAN ISO_GND strictly segreg
External Interfaces: DD-50 Sense Harness (J30), M8 High-Current Lug Terminals (J32-J34), Shunt Ke
Master Interconnect: AFE, Power Supply, MCU, Solid-State Switching, NTC Matrix, Balance Status, Isola

Open Source BMS Project

Sheet: /
File: 16s_BMS.kicad_sch

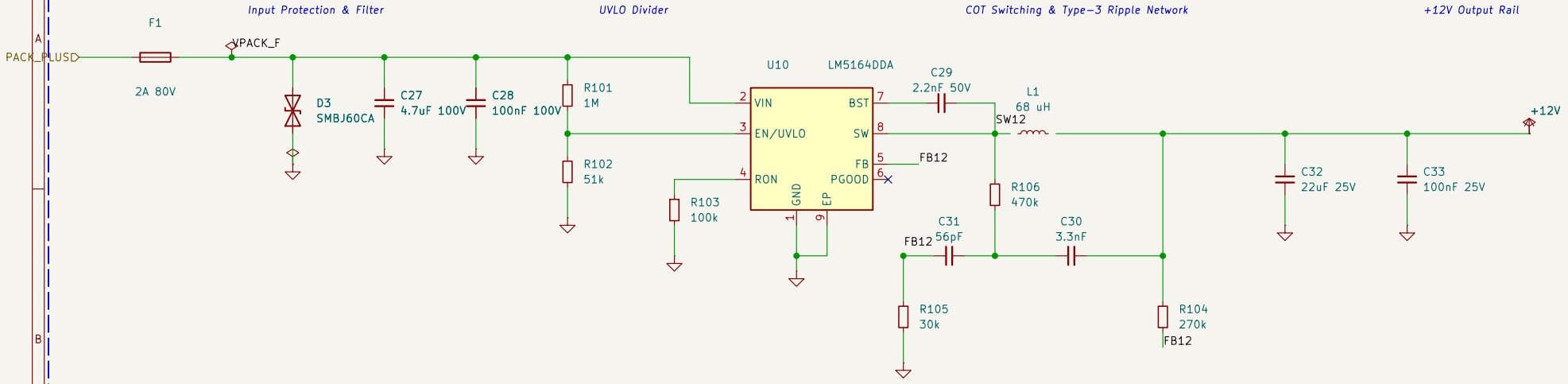
Title: 16S LiFePO4 Battery Management System - Master Top Level

Size: A3	Date: 2026-11-06	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/40

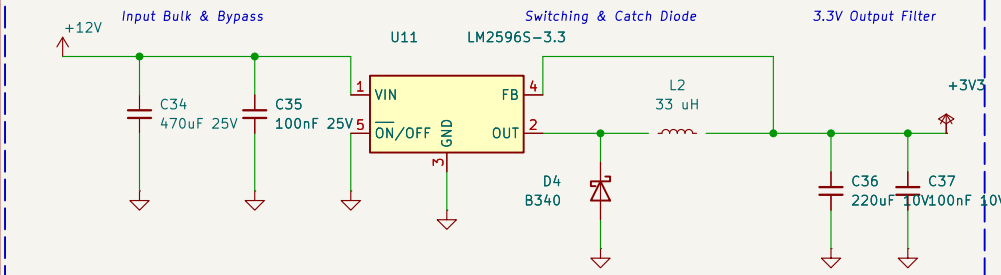


Sheet: /01_Cell_AFE/	
File: 01_Cell_AFE.kicad_sch	
Title: Analog Front End	
Size: A4	Date: 2026-10-16
KiCad E.D.A. 10.0.7	Rev: 1
	Id: 2/40

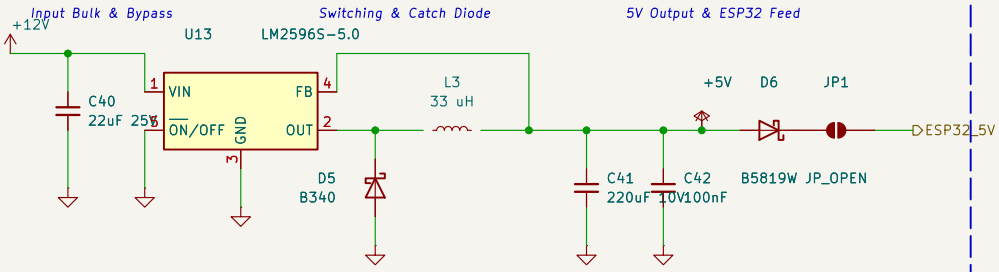
1. HIGH-VOLTAGE BUCK REGULATOR (LM5164): PACK+ (40V – 60V) -> +12V (1A)



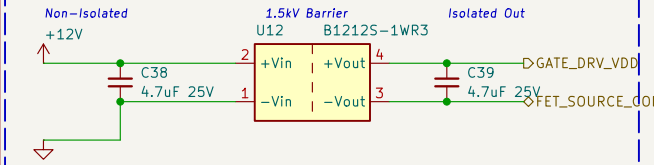
2. LOGIC POWER BUCK REGULATOR (LM2596S-3.3): +12V -> +3V3 (3A)



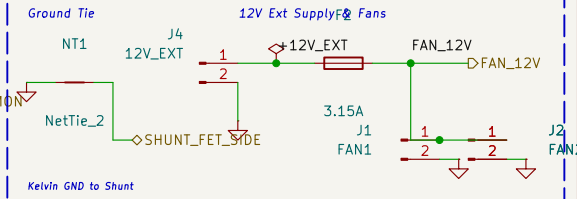
3. 5V SYSTEM POWER BUCK REGULATOR (LM2596S-5.0): +12V -> +5V (3A)



4. ISOLATED GATE DRIVE (B1212S)



5. FAN HEADERS & STAR GROUND TIE

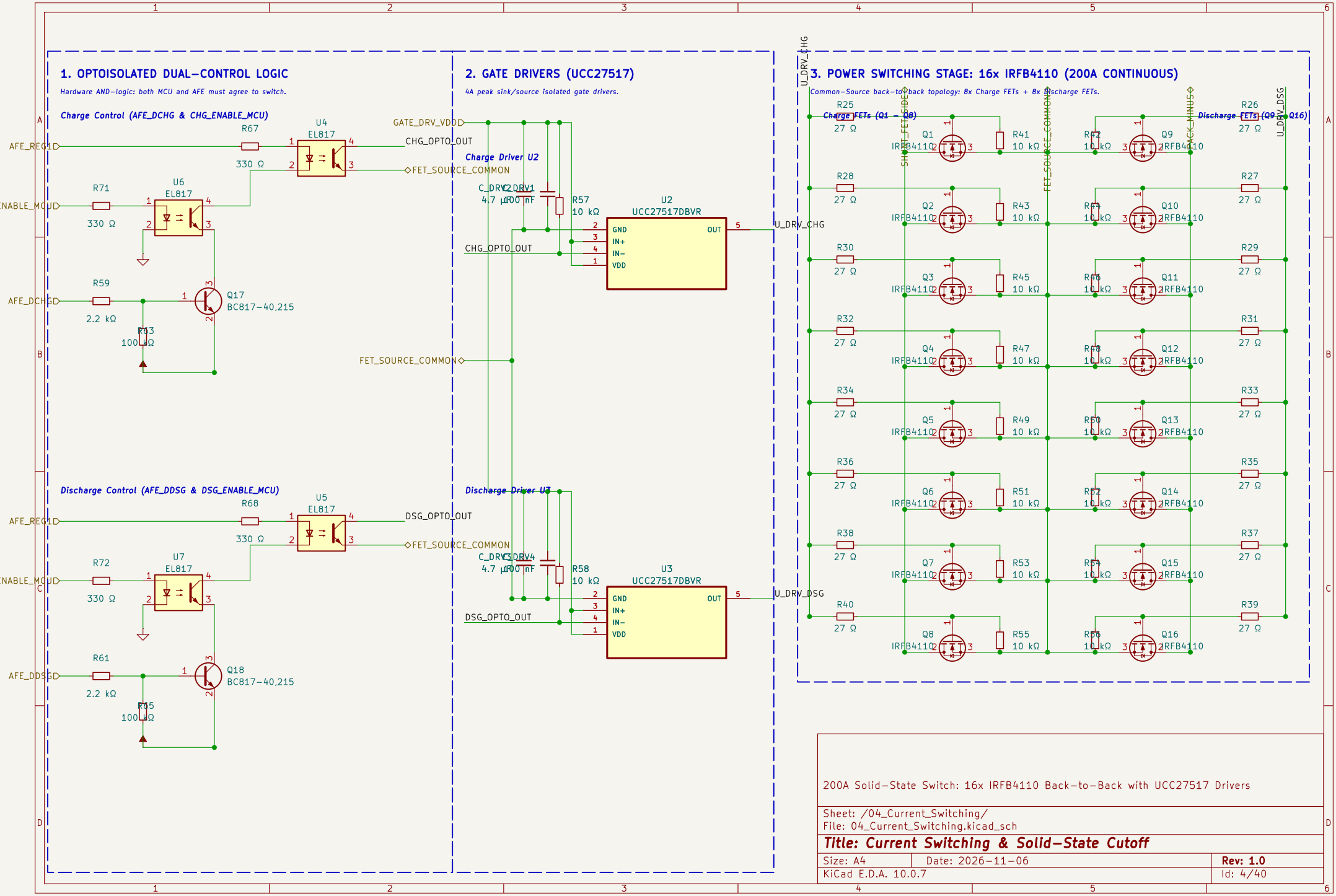


High-voltage buck (LM5164), 3.3V & 5V logic bucks (LM2596), isolated gate drive (B1212S)

Sheet: /02_Power_Supply/
File: 02_Power_Supply.kicad_sch

Title: 16S BMS Power Supply Subsystem

Size: A4	Date: 2026-11-06	Rev: 2.5
KiCad E.D.A. 10.0.7		Id: 3/40


1. OPTOISOLATED DUAL-CONTROL LOGIC
Hardware AND-logic: both MCU and AFE must agree to switch.
Charge Control (AFE_DCHG & CHG_ENABLE_MCU)
AFE_REG1D
R67
330 Ω
U4
EL817
GATE_DRV_VDD
CHG_OPTO_OUT
FET_SOURCE_COMMON
CHG_ENABLE_MCU
R71
330 Ω
U6
EL817
AFE_DCHGD
R59
2.2 kΩ
R63
100 kΩ
Q17
BC817-40,215
Discharge Control (AFE_DDSG & DSG_ENABLE_MCU)
AFE_REG1D
R68
330 Ω
U5
EL817
DSG_OPTO_OUT
FET_SOURCE_COMMON
DSG_ENABLE_MCU
R72
330 Ω
U7
EL817
AFE_DDSGD
R61
2.2 kΩ
R65
100 kΩ
Q18
BC817-40,215
2. GATE DRIVERS (UCC27517)
4A peak sink/source isolated gate drivers.
Charge Driver U2
C_DRV2_DRV1
4.7 µF
100 nF
R57
10 kΩ
U2
UCC27517DBVR
GND
IN+
IN−
VDD
OUT
CHG_OPTO_OUT
FET_SOURCE_COMMON
U_DRV_CHG
Discharge Driver U3
C_DRV3_DRV4
4.7 µF
100 nF
R58
10 kΩ
U3
UCC27517DBVR
DSG_OPTO_OUT
U_DRV_DSG
3. POWER SWITCHING STAGE: 16x IRFB4110 (200A CONTINUOUS)
Common-Source back-to-back topology: 8x Charge FETs + 8x Discharge FETs.
Charge FETs (Q1 – Q8)
Discharge FETs (Q9 – Q16)
SHORT_FET_GATES
FET_SOURCE_COMMON
BACK_MINUS
U_DRV_CHG
U_DRV_DSG
R25
27 Ω
R26
27 Ω
R28
27 Ω
R27
27 Ω
R30
27 Ω
R29
27 Ω
R32
27 Ω
R31
27 Ω
R34
27 Ω
R33
27 Ω
R36
27 Ω
R35
27 Ω
R38
27 Ω
R37
27 Ω
R40
27 Ω
R39
27 Ω
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
Q11
Q12
Q13
Q14
Q15
Q16
IRFB4110
R41
R43
R45
R47
R49
R51
R53
R55
R42
R44
R46
R48
R50
R52
R54
R56
10 kΩ
200A Solid-State Switch: 16x IRFB4110 Back-to-Back with UCC27517 Drivers
Sheet: /04_Current_Switching/
File: 04_Current_Switching.kicad_sch
Title: Current Switching & Solid-State Cutoff
Size: A4
Date: 2026-11-06
Rev: 1.0
KiCad E.D.A. 10.0.7
Id: 4/40

1. MCU POWER & DECOUPLING

C50 100nF
C51 100nF
C52 100nF
C54 100nF
C53 4.7uF

Decoupling: 4x 100nF (one per VDD pin) + 4.7uF bulk

FB10 120R/1A
C55 1uF
C56 10nF
VDDA

VDDA analog supply filtered via ferrite bead + 1uF/10nF

2. RESET, BOOT0 & HSE 8MHz OSCILLATOR

R120 10k
C59 100nF
MCU_NRST

JP10
BOOT0
R121 100k

JP10 open = BOOT from Flash (default); close to boot system ROM

Y1 8MHz
OSC_IN
OSC_OUT
C57 18pF
C58 18pF

Y1: 8 MHz HSE crystal with 18pF load capacitors

3. CH32V203C8T6 MCU CORE & PERIPHERALS

U20 CH32V203C8T6

Pin	Net
1 VBAT	
VDDA	VDDA
24 VDD_IO_1	
36 VDD_2	
48 VDD_IO_3	
8 VSSA	
23 VSS_1	
35 VSS_2	
47 VSS_3	
MCU_NRST 7 NRST	
BOOT0 44 BOOT0	
OSC_IN 5 OSC_IN	
OSC_OUT 6 OSC_OUT	
TEMP_MUX1_COM 14 PA4	
TEMP_MUX2_COM 15 PA5	
2 PC13	
3 PC14	
4 PC15	
BAL_SENSED 11 PA1	
ADDR0 16 PA6	
19 PB1	
20 PB2	
39 PB3	
40 PB4	
PA13 34	SWDIO
PA14 37	SWCLK
PA2 12	UART_TX
PA3 13	UART_RX
PB0 18	LED_STATUS
PB10 21	EE_SCL
PB11 22	EE_SDA
PB5 41	ADDR2
PB9 46	CAN_TX
PB8 45	CAN_RX
PB6 42	AFE_SCL
PB7 43	AFE_SDA
PA0 10	AFE_ALERT
PA8 29	CHG_ENABLE_MCU
PA9 30	DSG_ENABLE_MCU
PB12 25	TEMP_MUX_S0
PB13 26	TEMP_MUX_S1
PB14 27	TEMP_MUX_S2
PB15 28	TEMP_MUX_S3
PA7 17	ADDR1
PA10 31	
PA11 32	
PA12 33	
PA15 38	

4. PROGRAMMING & DEBUG INTERFACES

J10 WCH-LinkE
SWDIO
SWCLK
MCU_NRST

R125 1k
D10 GREEN
LED_STATUS

J11 Debug UART
UART_TX
UART_RX

5. I2C2 BUS: FRAM MEMORY & STATUS OLED DISPLAY

J12 OLED Display (I2C)
EE_SCL
EE_SDA

U21 FM24CL64B-G
C61 100nF
A0 A1 A2 GND VCC SDA SCL WP

R129 4.7k
R128 4.7k
R131 10k
JP14
EE_WP

J12 (OLED): 1:GND, 2:+3V3, 3:SCL, 4:SDA (SSD1306/SH1106)

JP14 open = Write Enabled (default); close to hardware write-protect

6. HARDWARE PACK ADDRESS & GATE DRIVER CONTROLS

JP11
ADDR0
ADDR1
ADDR2

CHG_ENABLE_MCU
R126 100k
R127 100k
DSG_ENABLE_MCU

CH32V203C8T6 BMS Main Controller with I2C2 FRAM and Balance Sense

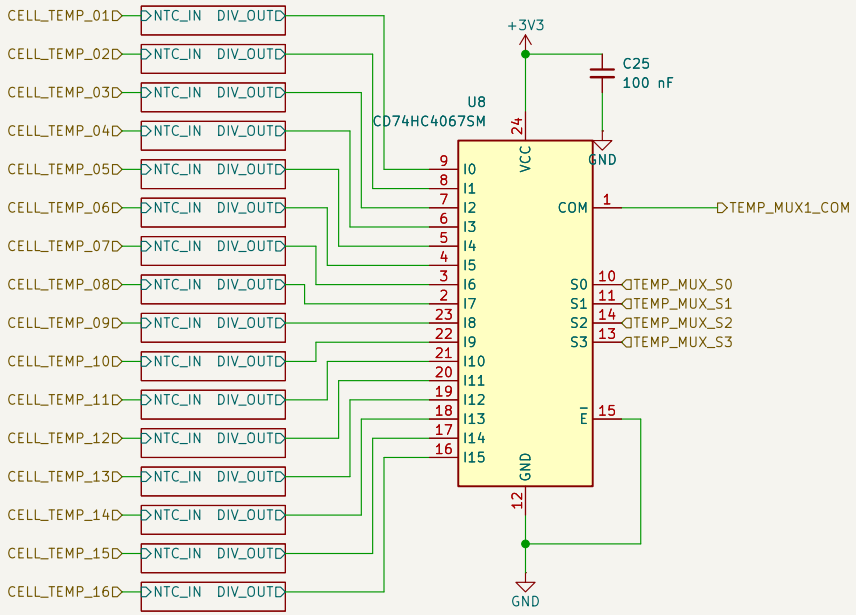
Sheet: /03_MCU_Control/
File: 03_MCU_Control.kicad_sch

Title: 03_MCU_Control

Size: A4	Date:	Rev: 4
KiCad E.D.A. 10.0.7		Id: 5/40

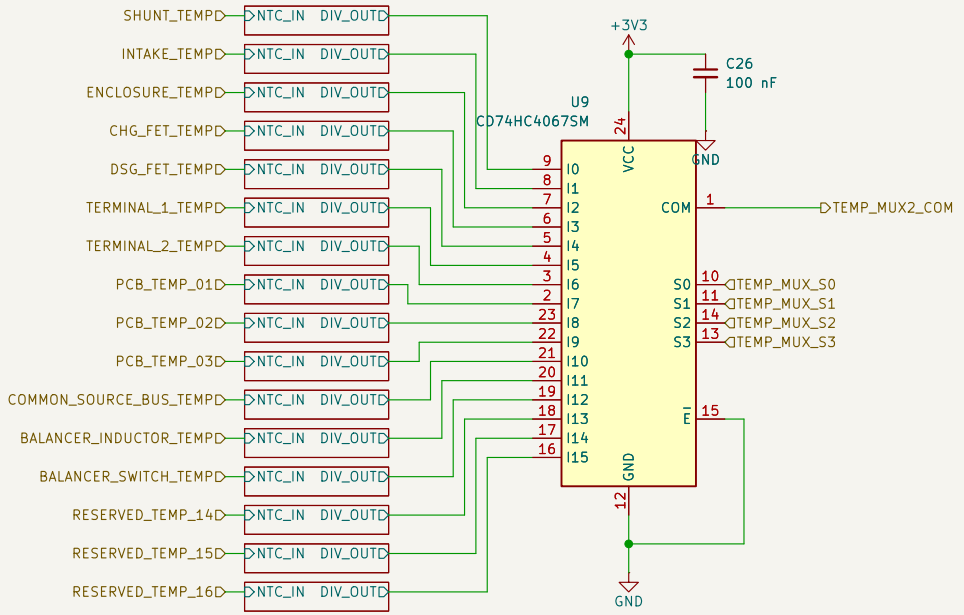
1. CELL TEMPERATURE SENSING (MUX 1 – CD74HC4067SM)

16 NTC Thermistor dividers for 16S battery cells.



2. SYSTEM & FET TEMPERATURE SENSING (MUX 2 – CD74HC4067SM)

16 NTC Thermistor dividers for shunt, FETs, inductors, and PCB.

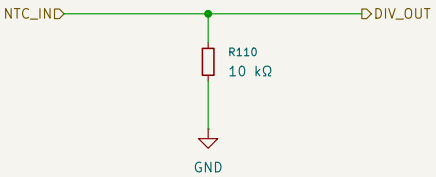


32-Channel Multiplexed NTC Matrix (Dual CD74HC4067)

Sheet: /05_Temperature_Sensing/
File: 05_Temperature_Sensing.kicad_sch

Title: Temperature Sensing & Thermal Management

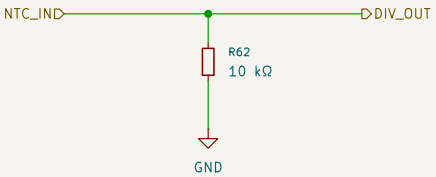
Size: A4	Date: 2026-11-06	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 6/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

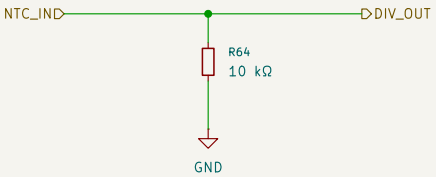
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 7/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER1/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

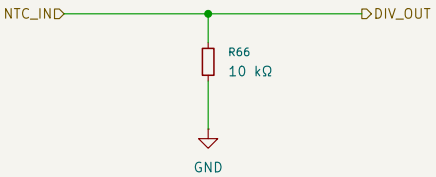
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 8/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER2/
File: 05_1_NTC_DIVIDER.kicad_sch

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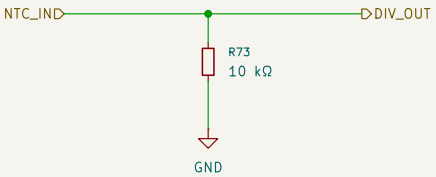
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 9/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER3/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

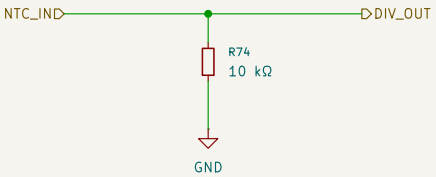
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 10/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER4/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

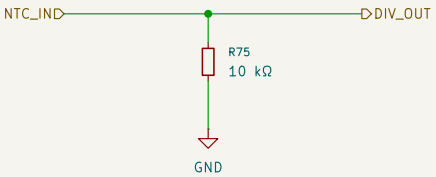
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 11/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER5/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

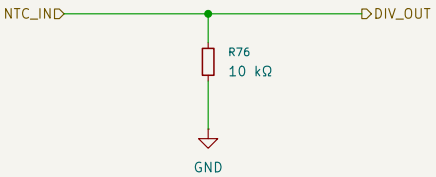
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KiCad E.D.A. 10.0.7		Id: 12/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER6/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

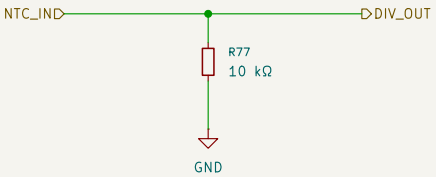
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 13/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER7/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

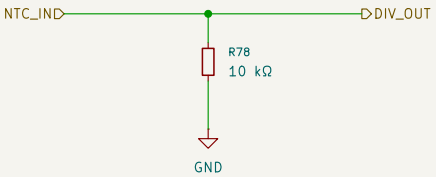
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 14/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER8/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

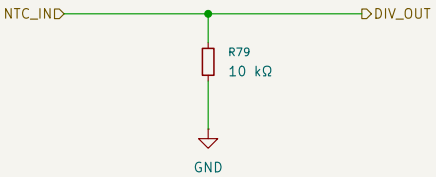
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 15/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER9/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

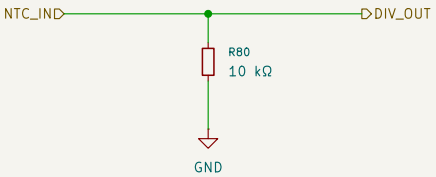
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 16/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER10/
File: 05_1_NTC_DIVIDER.kicad_sch

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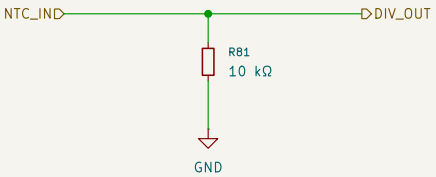
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 17/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER11/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

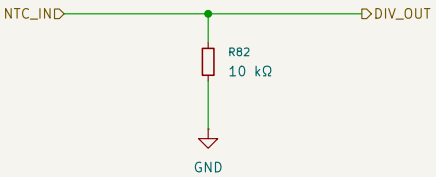
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 18/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER12/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

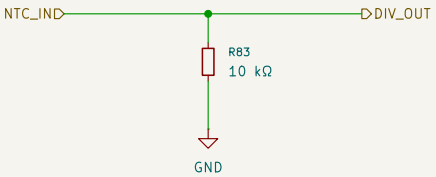
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 19/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER13/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

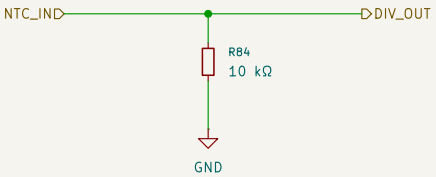
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 20/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER14/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

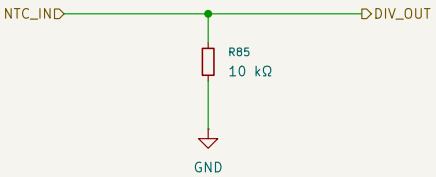
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 21/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER15/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

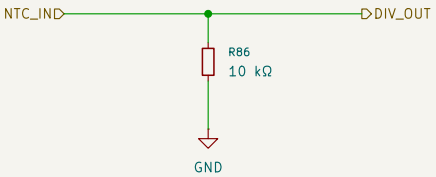
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 22/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER16/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

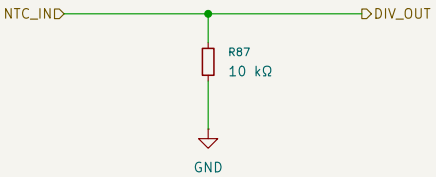
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 23/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER17/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

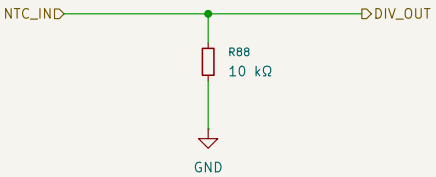
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 24/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER18/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

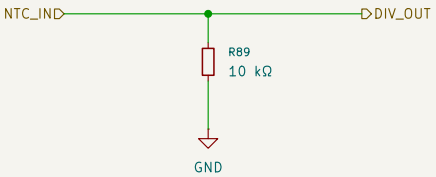
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 25/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER19/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

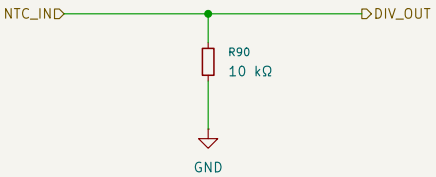
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 26/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER20/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

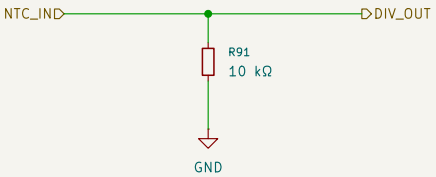
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 27/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER21/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

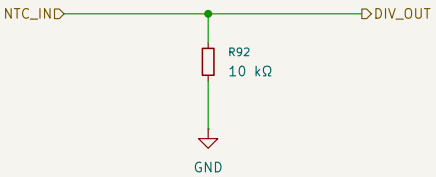
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 28/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER22/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

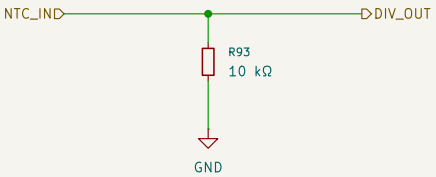
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 29/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER23/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

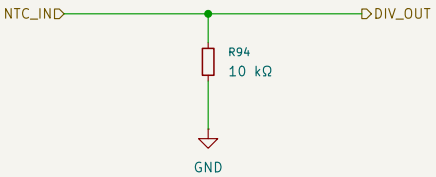
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 30/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER24/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

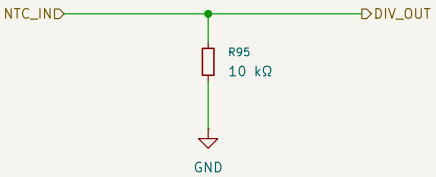
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 31/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER25/
File: 05_1_NTC_DIVIDER.kicad_sch

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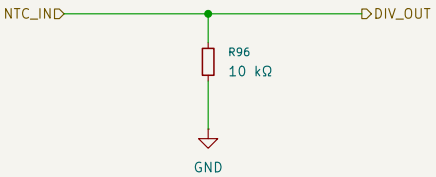
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 32/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER26/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

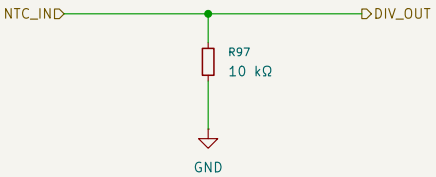
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 33/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER27/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

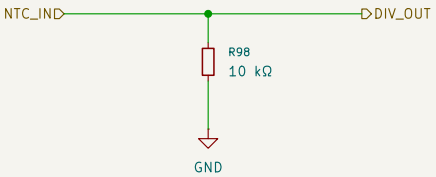
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 34/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER28/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

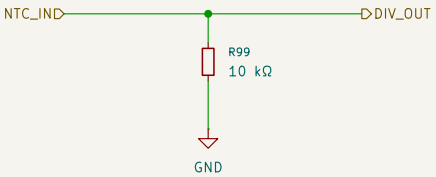
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 35/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER29/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

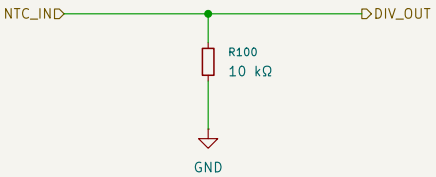
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 36/40



Sheet: /05_Temperature_Sensing/NTC_DIVIDER30/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 37/40



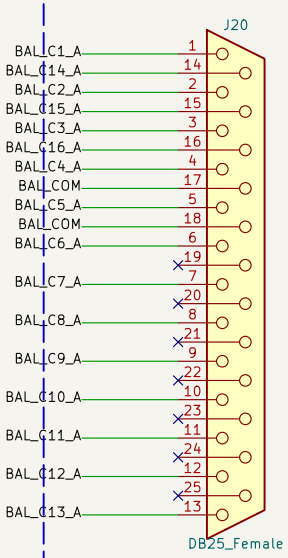
Sheet: /05_Temperature_Sensing/NTC_DIVIDER31/
File: 05_1_NTC_DIVIDER.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 38/40

1. BALANCER LED INPUT (J20)

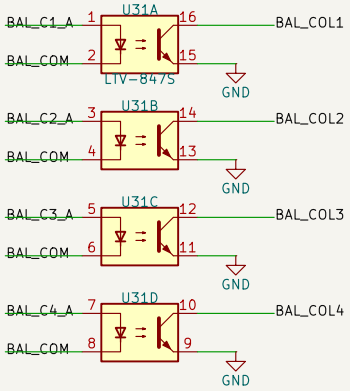
From ETA3000 Active Balancer LED pads. Galvanic isolation from BMS GND.



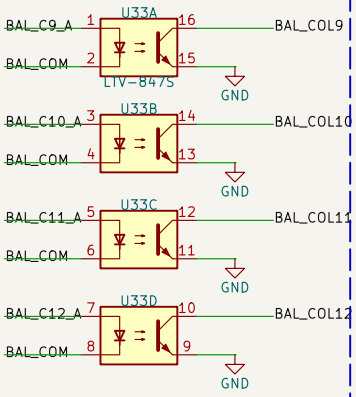
2. OPTOCOUPLER ISOLATION BARRIER (U31–U34)

LTV-847S Quad DC Optocouplers. Transistor outputs pulled to logic GND.

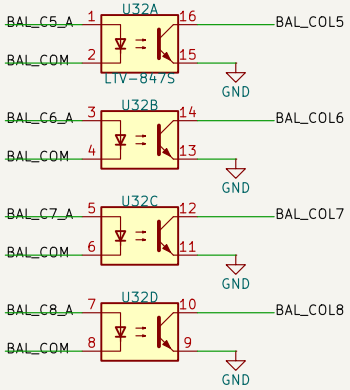
U31 (Channels 1..4)



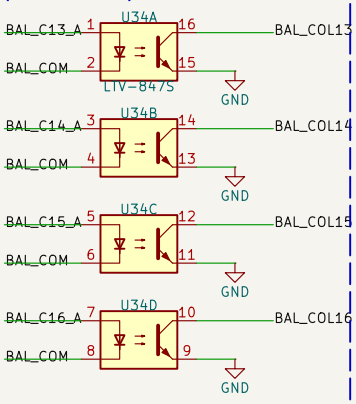
U33 (Channels 9..12)



U32 (Channels 5..8)

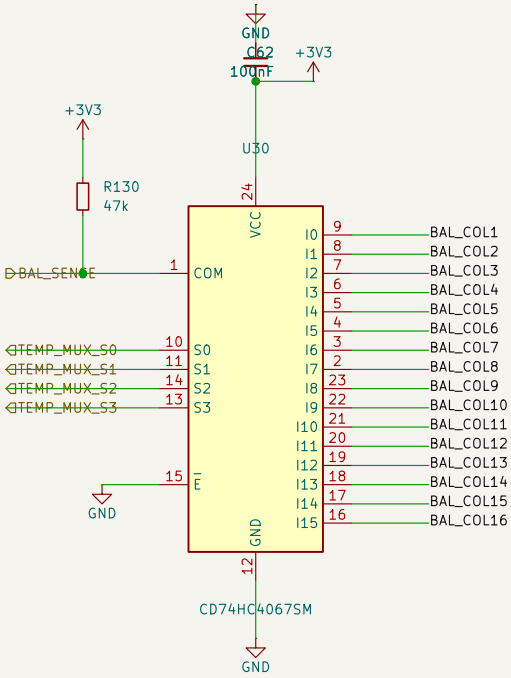


U34 (Channels 13..16)



3. ANALOG MUX & STATUS SENSE

CD74HC4067SM selects cell balance status. Shared select lines with temp mux.



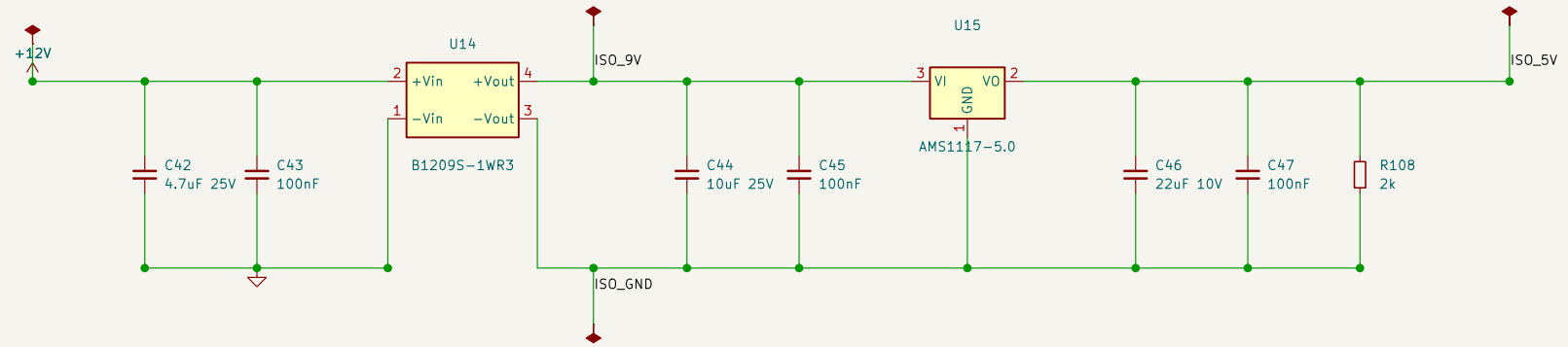
16S Active Balancer Optocoupler Status Interface

Sheet: /06_Balance_Status/
File: 06_Balance_Status.kicad_sch

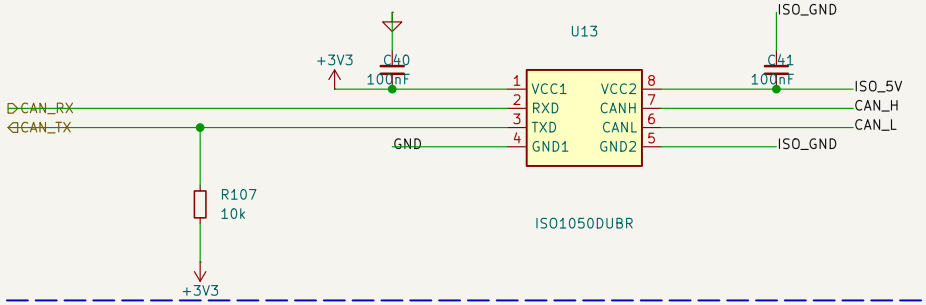
Title: 06_Balance_Status

Size: A4	Date:	Rev: 1
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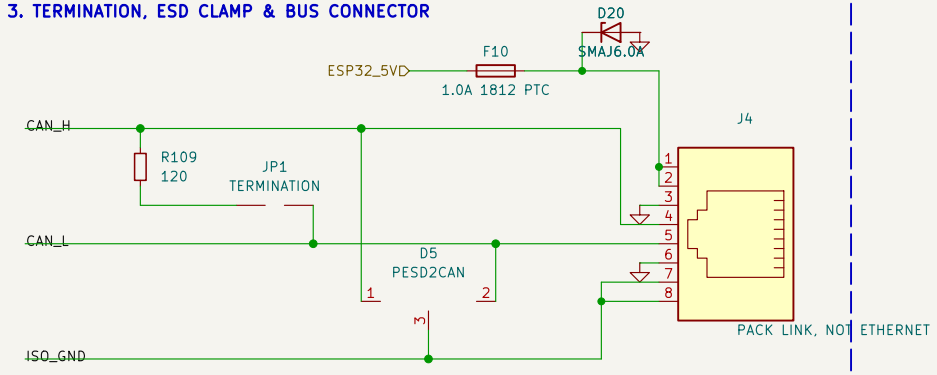
1. ISOLATED BUS POWER SUPPLY (+12V -> ISO_9V -> ISO_5V)



2. ISOLATED CAN TRANSCEIVER (ISO1050DUBR)



3. TERMINATION, ESD CLAMP & BUS CONNECTOR



Isolated CAN Transceiver (ISO1050), Isolated DC-DC (B1209S), LDO (AMS1117-5.0)

Sheet: /07_Comms/
File: 07_Comms.kicad_sch

Title: 16S BMS Isolated CAN Communication

Size: A4	Date: 2026-11-05	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 40/40